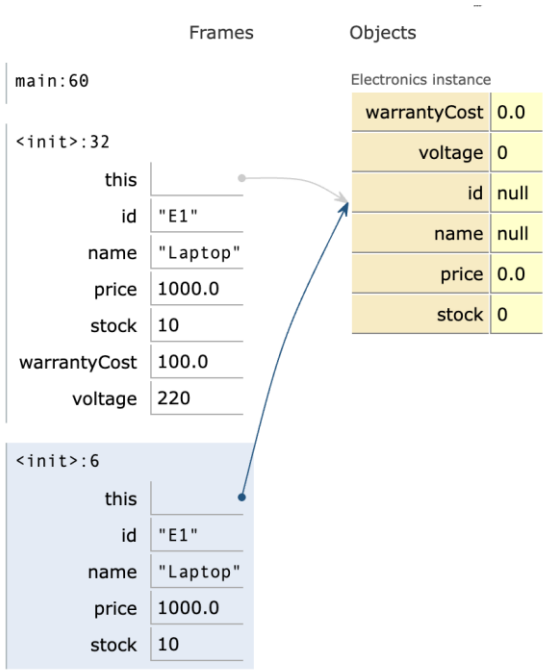
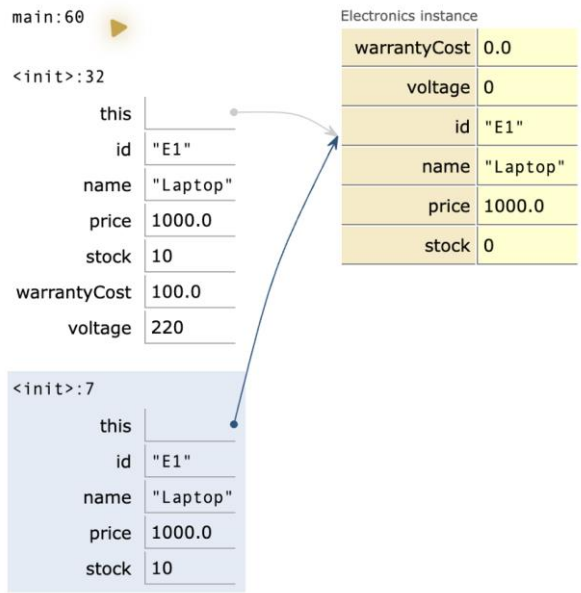
Java Tutorial

Objective: To analyze and explain the execution flow of the given program using python tutor, focusing on inheritance, constructor chaining with super(), and method overloading .

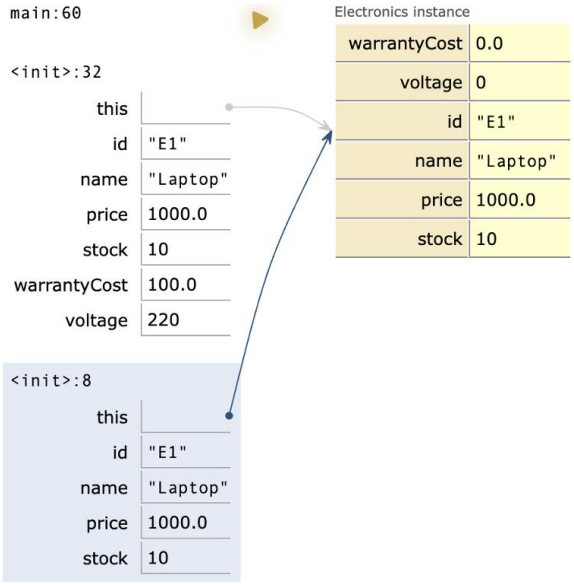


The parent constructor is initializing the Electronics object field by field, having already set id and name. Next, it will assign price and stack, followed by child-specific warrantyCost and voltage.

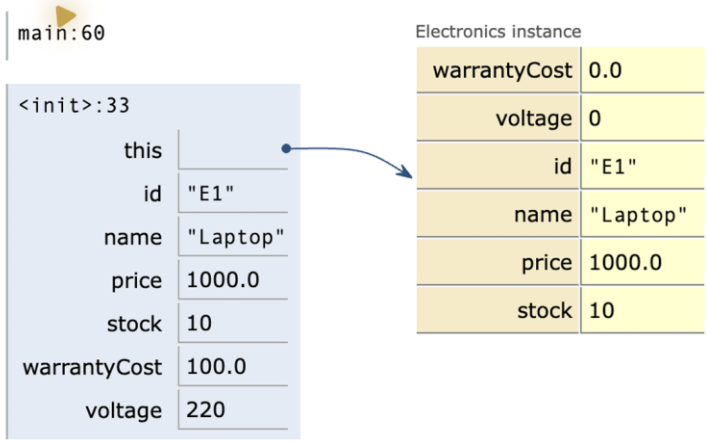


This shows java constructor chaining through inheritance. The Electronics subclass constructor (line 32) calls the superclass constructor (line 7) via super to set inherited fields

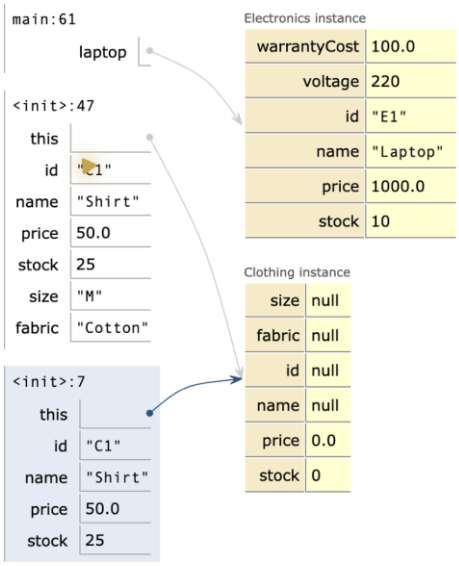
like id,name,price. Both stack frames point to the exact same Electronices object currently being built on the heap.



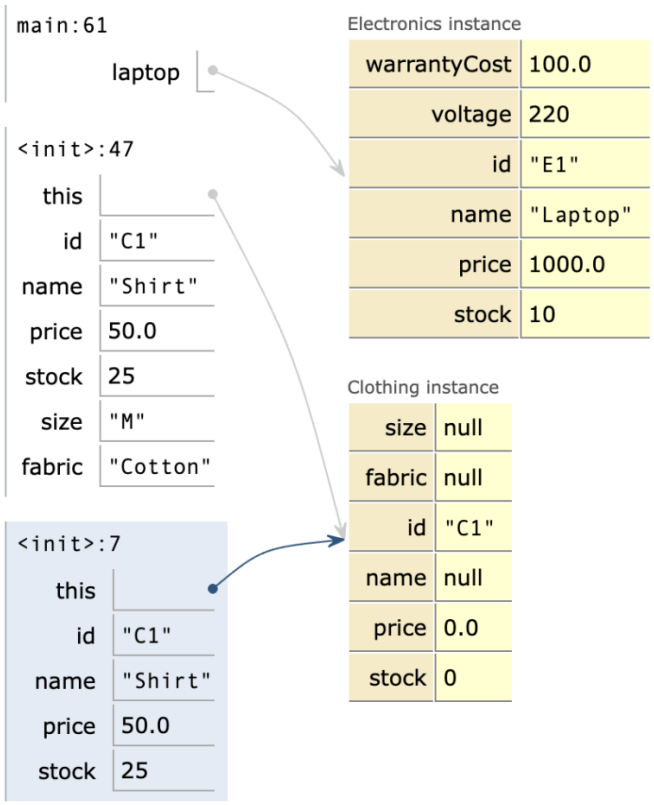
At line 8, superclass constructor finishes execution after setting the final field, completing all inherited properties. The control now returns to the subclass constructor (line 32) to assign its remaining specific field, warrantyCost and voltage.



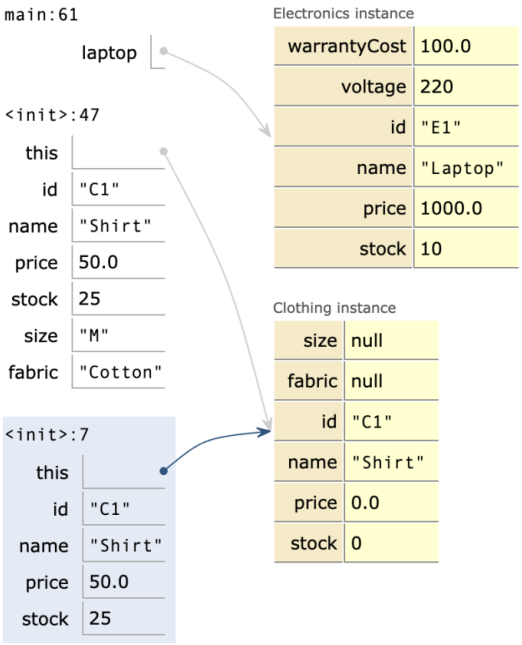
Control is now back in the subclass constructor at line 33 after the superclass constructor finished. With the base properties(id,name,stock)fully set,the code is ready is ready to initialize the subclass fields.



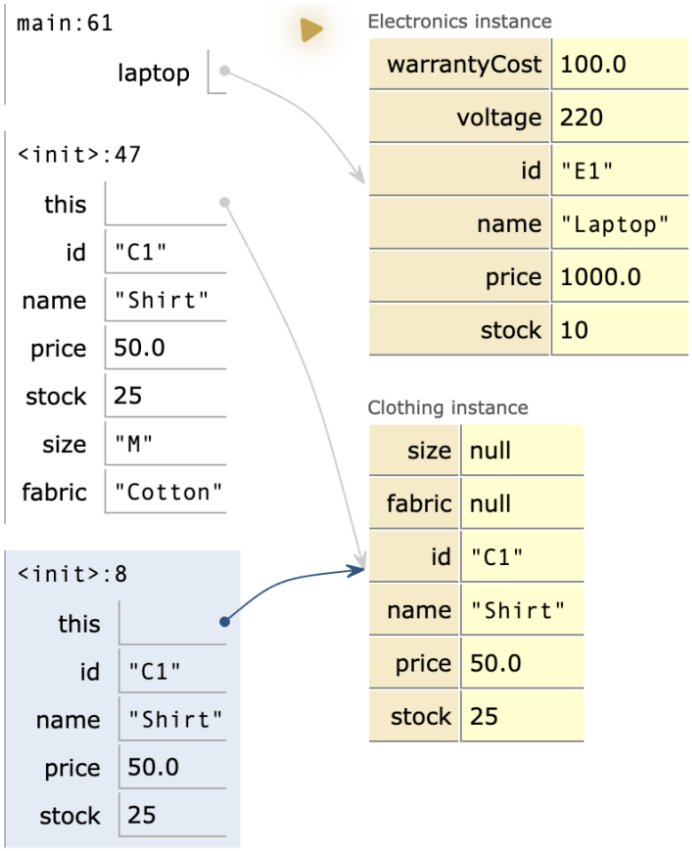
At line 61, a second subclass ‘Clothing’ is being created for “C1”,while the laptop object remains fully initialized. The clothing constructor(line 47) has called the parent constructor(line 7),which is currently running to set up the base fields for this new instance on the heap.



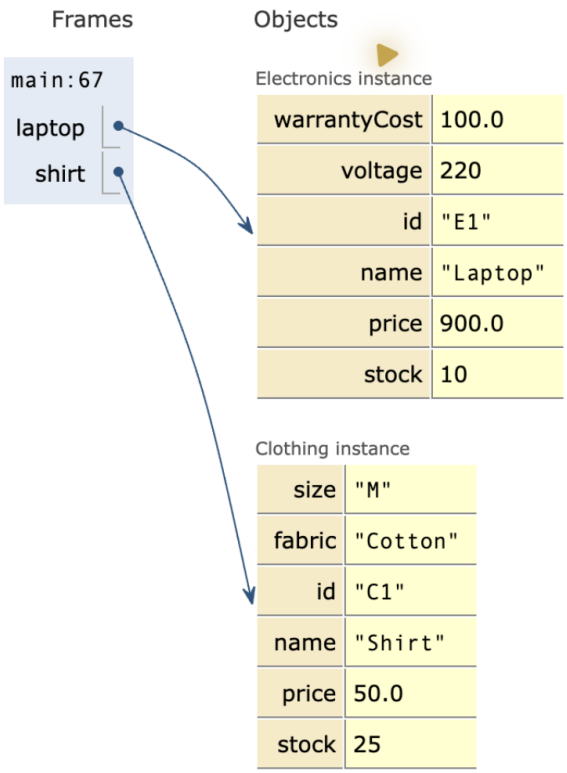
This image captures the first assignment which is executing inside the parent constructor for the newly allocated Clothing object. The statement this.id=id has completed updating the id attribute on the heap from null to c1.



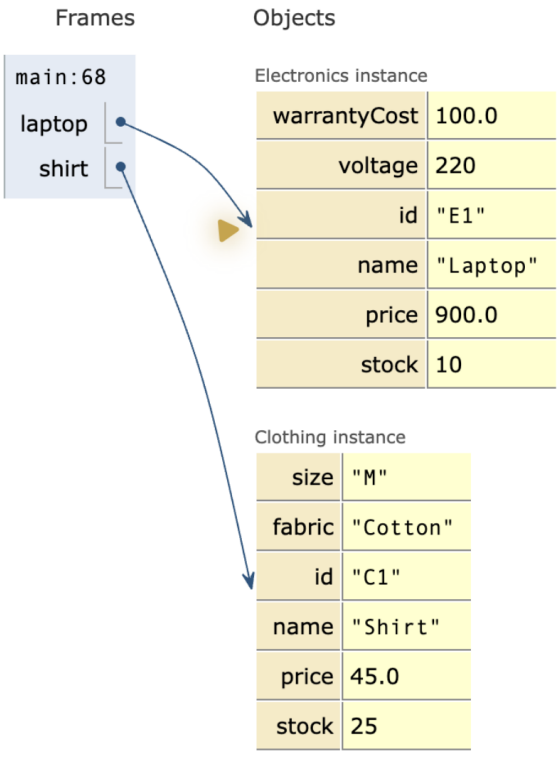
Here we see the second assignment executing inside the parent constructor for Clothing. Id and name are now assigned, while other base fields are yet to receive their values.



This image shows the completion of the parent constructor for the Clothing instance. Price and stock have executed. The parents stack is going to exit, returning execution to Clothing constructor to initialize size and fabric.



Here both objects are fully instantiated and actively referenced in the main method. Both constructors are completed and popped off the stack. The price is reduced from 1000 to 900.



In line 68 the shirt object has been discounted by 10% by dropping its price from 50.0 to 45.0 just like the laptop earlier. Both objects now show their discounted prices while being independently references by the laptop and shirt variables.

Thank you!!

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